Applied Economics Analysis

Final Project Presentation

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Motivation

- Higher income improves mental health.
- The relationship runs in **both directions:** Generates persistent effects on each other.
- Mental health measured years earlier predicts well-being more strongly than current income.

This Paper

Core Question

Does a **lifetime of high earnings** translate into better psychological well-being and life satisfaction **in later life**?

- I use **50+ years of longitudinal data** to trace individuals from early career to old age.
- I connect **lifetime labor income trajectories** to **psychological distress** and **life satisfaction** at age 50+.

Data: Panel Study of Income Dynamics (PSID)

- **Longest running longitudinal household survey** in the world.
- Annual waves 1968–1996, biennial from 1997.
- Tracks individuals and their descendants over time.
- **85,536 individuals**.

Main Variables:

Concept	Variable	Available Since
Hourly wage (real USD 2019)	`wage_real`	1968
Employment status	`empl_status`	1979
Psychological distress	K6 scale (0–24)	1997
Life satisfaction	1–5 scale	2007
Health status	1–5 scale	1994

Variable Construction

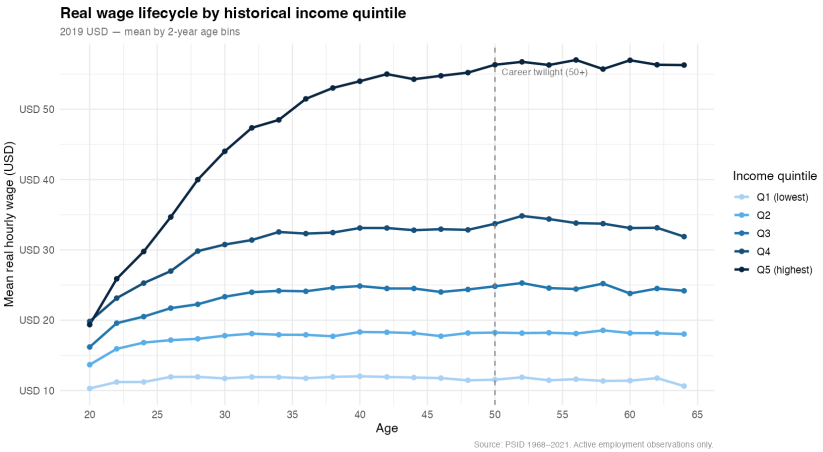
Income trajectory:

- Historical average real wage: mean hourly wage across all employed person-years (CPI-U, base 2019).
- Income quintiles: Q1 (lowest) to Q5 (highest), requires ≥ 3 years of observed wage.
- Employment indicator: equals 1 if individual reports working in that year, 0 otherwise.
- Coefficient of variation (CV): standard deviation divided by mean of employment status across all observed years (age 20–65).

Well-being outcomes (age 50+):

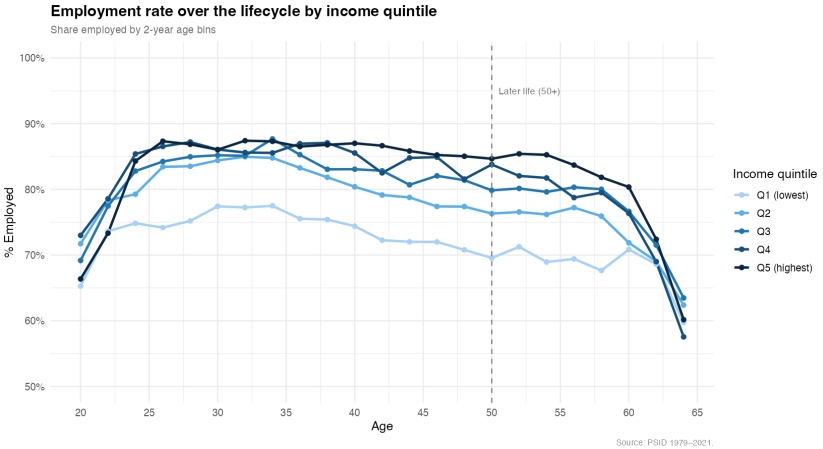
- **K6 distress**: Kessler-6 scale (0–24); scores ≥ 13 indicate clinically significant distress; available from 1997
- **Life satisfaction**: 1 = completely satisfied, 5 = not at all satisfied; available from 2007

Real Wage Lifecycle by Income Quintile



Mean real hourly wage by 2-year age bins. Active employment observations only.

Employment Rate Over the Lifecycle by Income Quintile



Share employed by 2-year age bins. PSID 1979–2021.

Labor Trajectory and Well-being at Age 50+

Quintile	CV Employment (20–65)	K6 (50+)	Life Sat. (50+)
Q1 (lowest)	0.545	3.32	2.19
Q2	0.411	3.03	2.18
Q3	0.380	2.51	2.11
Q4	0.354	2.41	2.05
Q5 (highest)	0.330	2.00	1.93

CV = coefficient of variation of employment status (lower = more stable). K6 and life satisfaction measured at age 50+.

Identification Strategy

Two baseline specifications, each estimated for K6 and life satisfaction:

(1) Earnings trajectory:

$$y_{it} = \alpha + \beta \cdot \bar{w}_i + \gamma X_{it} + \lambda_t + \varepsilon_{it}$$

(2) Employment stability:

$$y_{it} = \alpha + \beta \cdot CV_i + \gamma X_{it} + \lambda_t + \varepsilon_{it}$$

where X_{it} includes age, current employment status, and health status; λ_t are year fixed effects.

Threats to identification

- **Reverse causality:** poor mental health reduces earnings and employment stability
- **Selection:** healthier individuals may sort into high-earning careers from the start

Preliminary Results

	Dependent variable:			
	K6 Distress		Life Satisfaction	
	(1)	(2)	(3)	(4)
Mean real wage (hist.)	-0.012^{***}		-0.002^{***}	
	(0.002)		(0.0004)	
CV employment		0.337^{***}		0.025^{**}
		(0.059)		(0.011)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	37,639	37,639	37,639	37,639
R^2	0.098	0.098	0.101	0.099
Adjusted R^2	0.097	0.097	0.101	0.099
Note:	$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$			
	Robust SE clustered by individual in parentheses.			

Controls include age, employment status and health.

Conclusion

- High-earning individuals experience **lower psychological distress** throughout their working lives but this gap **narrows significantly after 50.**
- Those with higher lifetime earnings report **greater life satisfaction** in later life.

Next Steps

Establish whether these patterns reflect a **causal effect** of lifetime earnings or **selection** using individual fixed effects and, if possible, an IV or event-study approach around retirement.

Thank You

Questions?